INVERTIS UNIVERSITY BAREILLY
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BCS-251 – Programming for problem Solving Lab

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Laboratory Outcomes

- To formulate the algorithms for simple problems
- To translate given algorithms to a working and correct program
- To be able to correct syntax errors as reported by the compilers
- To be able to identify and correct logical errors encountered at run time
- To be able to write iterative as well as recursive programs
- To be able to represent data in arrays, strings and structures and manipulate them through a program
- To be able to declare pointers of different types and use them in defining self referential structures.
- To be able to create, read and write to and from simple text files.

Assignments for lab classes are as follows:

1: Problem solving using computers: Familiarization with programming environment

2: Variable types and type conversions: Simple computational problems using arithmetic expressions

3: Branching and logical expressions: Problems involving if-then-else structures

4: Loops, while and for loops: Iterative problems e.g., sum of series

5: 1D Arrays: searching, sorting: 1D Array manipulation

6: 2D arrays and Strings, memory structure: Matrix problems, String operations

7: Functions, call by value: Simple functions

8: Numerical methods (Root finding, numerical differentiation, numerical integration): Numerical methods problems

9: Recursion, structure of recursive calls: Recursive functions

10: Pointers, structures and dynamic memory allocation: Pointers and structures

11: File handling: File operations